IBM DATA SCIENCE PROFESSIONAL CERTIFICATE — CAPSTONE PROJECT

Winning Space Race with Data Science

Predicting SpaceX Falcon 9 First Stage Landing Success for Commercial Launch Bidding

Github link to notebooks: https://github.com/marvinminhaz/IBM-Data-Science-Capstone-SpaceX/tree/main/notebooks

Author: Marvin | **Date:** November 2026
Github: github.com/marvinminhaz

Presentation Outline

Section	Topics Covered in Template
1. Executive Summary	High-level summary of problem, methodology, and core results
2. Introduction	SpaceX Falcon 9 reusability ($62M vs. $165M) & Space Y business goal
3. Methodology	Data Collection (API & Web Scraping), Wrangling, EDA, and ML flowcharts
4. EDA & Visual Results	6 Seaborn Scatter/Bar/Line plots & 10 SQL Database Queries
5. Interactive Analytics	Folium Geospatial Proximity Maps & Plotly Dash Dashboard Analytics
6. Predictive Analysis (ML)	SVM, Logistic Regression, Decision Tree, KNN & Confusion Matrix
7. Appendix	Complete results for all 10 SQL queries from the IBM lab

Executive Summary

1. Problem & Data Pipeline

- **Objective:** Predict Falcon 9 first-stage recovery to help competitor "Space Y" bid against SpaceX ($62M vs $165M).
- **Data Collection and Wrangling:** Extracted 90 Falcon 9 launch records via static CSV datasets.
- **Labelling and Feature Enginneering:** Converted mission outcomes into a binary target Class (1 for successful landing, 0 for failure, yielding a 66.67% baseline success rate)

2. Key Exploratory Insights

- **Time & Experience:** Landing success jumped from 0% (2010) to 90% (2019–2020) as flight numbers increased.
- **Sites & Payloads:** CCAFS SLC 40 had the most launches (55), KSC LC 39A had the highest success rate, and heavy payloads (~15,600 kg) landed with near-perfect reliability
- **Orbits:** VLEO achieved high success (~85.7%) across 14 flights, while high-volume GTO (27) and ISS (21) had lower rates (~52% and ~62%).

3. Geospatial & Dashboard Findings

- **Folium Map:** All launch pads sit within 0.87–1.2 km of coastlines and railways, but >15 km from major cities for safety.
- **Plotly Dash:** Interactive filtering confirms Block 5 boosters carry up to 15,600 kg while maintaining reliable drone-ship recovery.

4. Machine Learning Results

- **Top Models:** Logistic Regression, SVM, and KNN tied at **83.33% test accuracy** on the 18-sample holdout set.
- **Recommendation:** Deploy Logistic Regression/SVM (100% recall, 0 false negatives) for automated mission cost estimation.

Introduction & Business Problem

Project Background

- **Cost Disruption:** SpaceX advertises Falcon 9 rocket launches on its website at **$62 million**, whereas other providers cost upward of **$165 million** per launch.
- **First-Stage Reusability:** Much of the savings comes from SpaceX's ability to land and reuse the first stage (which accounts for ~70% of launch cost) on ground pads (RTLS) or drone ships (ASDS).
- **Competitor Goal (Space Y):** As a data scientist at rival startup *Space Y*, our goal is to build a machine learning pipeline that predicts whether SpaceX will successfully recover the first stage for any given mission profile.
- **Bidding Application:** Knowing whether the booster will be reused allows Space Y to accurately infer SpaceX's marginal launch cost and submit competitive contract bids.

COMMERCIAL LAUNCH COST COMPARISON

$62M

SpaceX Falcon 9 (Reusable Stage 1)

$165M+

Traditional Providers (Expendable)

SECTION 01

Methodology

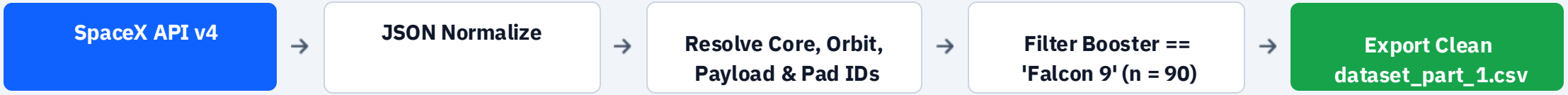
Data Collection, Data Wrangling, Exploratory Data Analysis, and Predictive Modeling Architecture

Github: https://github.com/marvinminhaz/IBM-Data-Science-Capstone-SpaceX

Methodology: Data Collection & Flowchart

- **Primary Data Source:** The data collection methodology utilized the official SpaceX REST API to programmatically request and parse comprehensive launch records, rocket configurations, and flight metrics via HTTP GET requests
- **Fallback Data Pipeline:** Because the original SpaceX API endpoint was taken down, the pipeline pivoted to using a static JSON dataset (API_call_spacex_api.json) provided by the IBM Skills Network capstone project, alongside pre-processed CSV datasets (dataset_part_1.csv).
- **Primary Data Source:** The data collection methodology utilized the official SpaceX REST API to programmatically request and parse comprehensive launch records, rocket configurations, and flight metrics via HTTP GET requests
- **Github**: https://github.com/marvinminhaz/IBM-Data-Science-Capstone-SpaceX/blob/main/notebooks/1.%20data-collection.ipynb

DATA COLLECTION FLOWCHART (TEMPLATE DIAGRAM)



Methodology: Data Wrangling & Exploratory Analysis

Github - https://github.com/marvinminhaz/IBM-Data-Science-Capstone-SpaceX/blob/main/notebooks/2.%20data-wrangling.ipynb

Data Wrangling Pipeline

- **Missing Value Imputation:** Replaced NaN entries in PayloadMass with the column mean (**6,123.55 kg**); retained LandingPad nulls where drone ships/ground pads were not used.
- **Target Label Engineering (Class):** Mapped 8 raw Outcome types into binary label:
  • Class = 1 (60 launches, 66.7%): True ASDS, True RTLS, True Ocean
  • Class = 0 (30 launches, 33.3%): False ASDS, False RTLS, False Ocean, None ASDS, None None
- **Feature One-Hot Encoding:** Applied pd.get_dummies() to Orbit, LaunchSite, LandingPad, and Serial, producing **83 total numerical features** (`dataset_part_3.csv`).

EDA & Interactive Visual Workflow

- **Statistical Visualization:** Plotted multivariate relationships using Matplotlib and Seaborn (catplot, barplot, lineplot) to isolate flight number, payload, and orbit boundaries.
- **Relational SQL Queries:** Loaded `Spacex.csv` (101 rows) into SQLite to run 10 structured SQL queries aggregating payloads, launch counts, and booster outcomes.
- **Geospatial & Dash Analytics:** Built interactive Folium map layers (MarkerCluster, Haversine distance polylines) and a real-time Plotly Dash web application with dropdowns and range sliders.

Methodology: Predictive Machine Learning Pipeline

1. Preprocessing & Split

Standardized all 83 features via preprocessing.StandardScaler(). Split dataset 80/20 using train_test_split(test_size=0.2, random_state=2) → **72 Training** & **18 Test samples**.

2. Hyperparameter Tuning

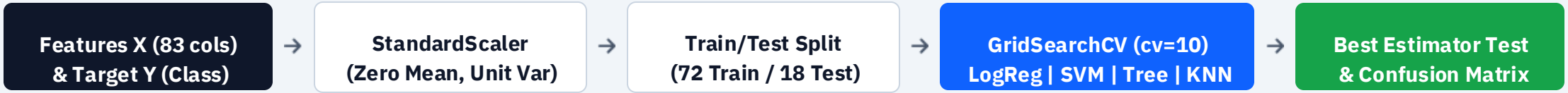
Trained 4 classifiers (Logistic Regression, SVM, Decision Tree, KNN) using GridSearchCV(cv=10) to identify optimal hyperparameter combinations without data leakage.

3. Model Evaluation

Evaluated models on the unseen 18-sample test set via accuracy scores, F1-score, recall, and confusion_matrix analysis to assess false positive risk.

Github: https://github.com/marvinminhaz/IBM-Data-Science-Capstone-SpaceX/blob/main/notebooks/6.%20predictive-modelling.ipynb

PREDICTIVE MODELING FLOWCHART (TEMPLATE DIAGRAM)



EDA with Visualization: Flight Number, Launch Site & Payload

1. Flight No. vs. Launch Site

KSC LC-39A (Flights 27–90)	**76.9% Success**
VAFB SLC-4E (Flights 4–65)	**76.9% Success**
CCAFS SLC-40 (Flights 1–90)	**60.0% Success**

Early missions (Flights 1–20) launched primarily at CCAFS SLC-40 with high failure rates. Flights > 40 show dense clusters of successful landings (Class 1) across all pads.

2. Payload vs. Launch Site

VAFB SLC-4E:	Max 9,600 kg (Polar)
KSC LC-39A:	Up to 15,600 kg
CCAFS SLC-40:	Up to 15,600 kg

For payloads > 10,000 kg, launches exclusively use KSC LC-39A and CCAFS SLC-40. VAFB SLC-4E never launches heavy payloads > 10,000 kg due to polar orbit constraints.

3. Flight No. vs. Orbit Type

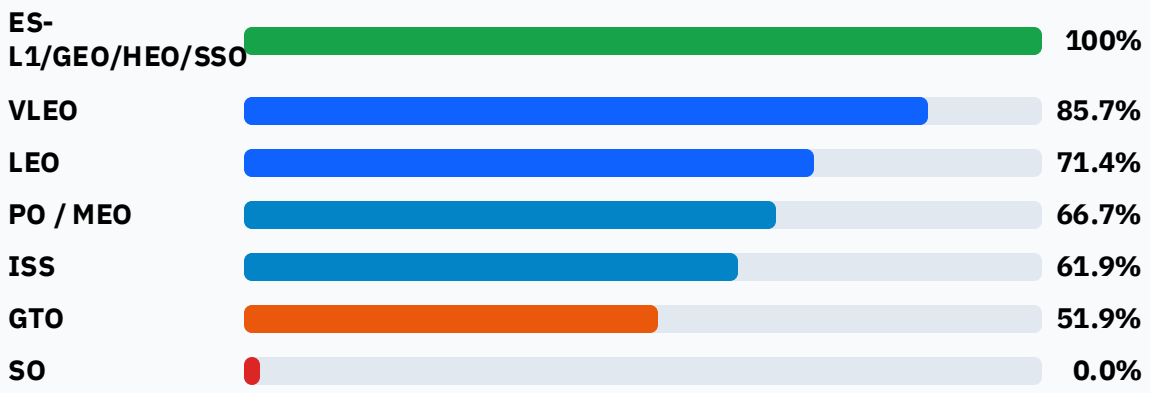
LEO / ISS:	**Consistent Success**
GTO Orbit:	**Mixed (51.9% Success)**
VLEO (Flights 64+):	**85.7% Success**

In LEO orbit, success correlates strongly with flight number. GTO remains challenging even at higher flight numbers due to high orbital velocity requirements.

Github: https://github.com/marvinminhaz/IBM-Data-Science-Capstone-SpaceX/blob/main/notebooks/3.%20data-visualization-and-feature-engineering.ipynb

EDA with Visualization: Orbit Success Rates & Yearly Trend

4. Success Rate by Orbit Type (Bar Chart)



5. Payload vs. Orbit Type

Heavy payloads enjoy high landing success in Polar (PO), LEO, and ISS orbits, whereas GTO displays mixed recovery outcomes across 3,000–6,500 kg payloads.

6. Launch Success Yearly Trend (2010–2020)

2010–12: 0% **2014:** 33% **2016:** 62% **2017: 83%** **2019: 90%** **2020: 84%**

First stage recovery rose sharply after 2015, stabilizing above 80% following the introduction of Falcon 9 Full Thrust and Block 5.

Github: https://github.com/marvinminhaz/IBM-Data-Science-Capstone-SpaceX/blob/main/notebooks/3.%20data-visualization-and-feature-engineering.ipynb

EDA with SQL: Summary of All 10 Required Lab Queries

#	SQL Task Prompt (Template Slides 19–28)	Core SQL Clause	Exact Query Output
Q1	Unique launch sites in space missions	`SELECT DISTINCT Launch_Site`	**CCAFS LC-40, VAFB SLC-4E, KSC LC-39A, CCAFS SLC-40**
Q2	5 records where launch sites begin with 'CCA'	`WHERE Launch_Site LIKE 'CCA%' LIMIT 5`	**5 rows returned (Flights 1–5 at CCAFS LC-40)**
Q3	Total payload mass carried by NASA (CRS)	`SUM(PAYLOAD_MASS__KG_) WHERE Customer='NASA (CRS)'`	**45,596 kg**
Q4	Average payload mass for booster F9 v1.1	`AVG(PAYLOAD_MASS__KG_) WHERE Booster='F9 v1.1'`	**2,928.4 kg (exact match)**
Q5	Date of first successful ground pad landing	`MIN(Date) WHERE Landing_Outcome='Success (ground pad)'`	**2015-12-22 (Flight 20, Orbcomm-OG2)**
Q6	Boosters with drone ship success & 4000–6000 kg payload	`WHERE Outcome='Success (drone ship)' AND Mass 4000..6000`	**F9 FT B1022, F9 FT B1026, F9 FT B1021.2, F9 FT B1031.2**
Q7	Total number of successful and failure mission outcomes	`SELECT Mission_Outcome, COUNT(*) GROUP BY 1`	**100 Successes (incl. 1 payload defect), 1 Failure (CRS-7)**
Q8	Boosters carrying maximum payload mass (subquery)	`WHERE Mass = (SELECT MAX(PAYLOAD_MASS__KG_))`	**12 Block 5 Boosters at 15,600 kg (B1048.4 to B1060.3)**
Q9	2015 failed drone ship landings with month names	`WHERE substr(Date,1,4)='2015' AND Outcome='Failure (drone ship)'`	**Jan 2015 (F9 v1.1 B1012) & Apr 2015 (F9 v1.1 B1015) at CCAFS**
Q10	Rank landing outcomes between 2010-06-04 and 2017-03-20	`GROUP BY Landing_Outcome ORDER BY COUNT(*) DESC`	**No attempt (10), Failure drone (5), Success drone (5), Ground (3)**

Interactive Visual Analytics: Folium Geospatial Analysis

3 Required Folium Map Layers

- **1. Global Launch Sites Map:** Plotted folium.Circle and folium.Marker objects at CCAFS LC-40 (28.562°N), CCAFS SLC-40 (28.563°N), KSC LC-39A (28.573°N), and VAFB SLC-4E (34.633°N) — all on coastal corridors for orbital inclination & over-ocean flight safety.
- **2. Color-Coded MarkerCluster:** Added MarkerCluster() with green markers (Class = 1) and red markers (Class = 0), highlighting KSC LC-39A's superior success ratio.
- **3. Haversine Proximity Polylines:** Calculated exact geodesic distances from launch pads to nearby infrastructure using the Haversine formula.

Proximity Target (CCAFS SLC-40)	Distance	Operational Reason
Closest Coastline	**0.87 km**	Safe ocean overflight & drone ship recovery
Closest Railway (NASA Rail)	**1.28 km**	Heavy rocket & solid propellant freight
Closest Highway (Pkwy)	**0.56 km**	Crew, payload & mobile crane transport
Nearest City (Titusville/Melbourne)	**16.3 – 54.6 km**	Blast radius & public safety buffer zone

Interactive Visual Analytics: Plotly Dash Dashboard

1. All Sites Success Pie Chart

Breakdown of total successful landings across all 4 launch sites:

KSC LC-39A: 41.7% **CCAFS LC-40: 29.2%** **VAFB: 16.7%** **SLC-40: 12.5%**

2. Highest Success Ratio Launch Site (KSC LC-39A)

Filtering dropdown to KSC LC-39A reveals the highest site efficiency:

Success (Class 1): 76.9% (10) **Failed (Class 0): 23.1% (3)**

3. Payload vs. Launch Outcome Scatter (All Range)

Color-coded by Booster Version Category (v1.0, v1.1, FT, B4, B5). Shows FT and B5 boosters dominate successful recoveries across all payload brackets up to 15,600 kg.

4. RangeSlider Filter Findings (0 – 10,000 kg)

Narrowing the dcc.RangeSlider reveals the highest landing success rate occurs in the **2,000 kg – 5,500 kg** payload band, while 0–2,000 kg and 6,000–7,000 kg bands show lower recovery rates.

Predictive Analysis: Model Accuracy & Confusion Matrix

Classification Accuracy Comparison

Algorithm	Best Hyperparameters	CV Score	Test Acc
Logistic Regression	`C=0.01, penalty='l2', lbfgs`	**84.64%**	**83.33% (15/18)**
Support Vector Machine	`kernel='sigmoid', C=1.0, γ=0.0316`	**84.82%**	**83.33% (15/18)**
K-Nearest Neighbors	`n_neighbors=10, p=1, auto`	**84.82%**	**83.33% (15/18)**
Decision Tree	`criterion='entropy', depth=4..8`	**87.68%**	**83.33% / 72.22%***

*Note: Decision Tree test accuracy varies between 72.22% and 83.33% due to random feature tie-breaking (`max_features='sqrt'`). Logistic Regression, SVM, and KNN deterministically achieve 83.33%.

Confusion Matrix (n = 18 Test Set)

TN = 3 True Negative (Not Landed)	**FP = 3** False Positive (Type I Error)
FN = 0 False Negative (Type II Error)	**TP = 12** True Positive (Landed)

Key Finding: 100% Recall on positive landings (12/12) with 3 false positives where failed historical landings were predicted as successful.

SECTION 04 — FINAL CONCLUSIONS

Key Conclusions & Bidding Strategy for Space Y

1. Booster Generation & Flight Maturity Dominate

Landing success increased from 0% (2010–2013) to >84% after 2017. Block 5 boosters operating above Flight #40 achieve near-100% recovery across LEO, ISS, and VLEO orbits.

2. Launch Site & Orbit Risk Profiles

KSC LC-39A yields the highest site success rate (76.9%). Orbits ES-L1, GEO, HEO, and SSO hold 100% recovery records, whereas GTO missions remain high-risk (51.9% recovery).

3. Recommended ML Classifier

Logistic Regression, SVM, and KNN all achieve 83.33% test accuracy. Logistic Regression & SVM are recommended for production deployment due to stability and calibrated probabilities.

4. Commercial Bidding Action Plan

When our model predicts Class = 1, SpaceX's cost is ~$62M (Space Y must bid aggressively). For heavy GTO missions predicted as Class = 0, SpaceX faces higher expendable costs, allowing Space Y higher margin bids.